Comparative Study of Hourglass and SimpleBaseline on MPII with YOLO Detector-based Fairness Protocol

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Abstract

We return to the empirical comparison of single-person pose estimators—Hourglass and SimpleBaseline—trained on MPII, versus a multi-person detector–pose pipeline YOLOv8-pose pre-trained on COCO keypoints. One common pitfall is evaluation mismatch*: single-person models are trained on person crops, and detector-based models run on full images with more than one person. We propose a low cost* detector-based fairness protocol *using IoU that selects, to the ground truth (GT) person, a single person box, and evaluates each method under the same single human input condition. Using PCKh@0.5 on MPII, we report findings with or without the protocol, identify failure cases—for example when the detector misses the GT person and assigns a pose to another individual—and discuss the causes of the lack of improvements in SimpleBaseline (in our runs). The protocol assists in avoiding false claims during cross-paradigm pose comparisons.*

1. Introduction

Human Pose Estimation (HPE) is the foundation of activity understanding, HCI, sports analytics. Stacked Hourglass [2] and SimpleBaseline [3], which are two of the most popular single-person baselines, were both studied on MPII [1]. On the other hand, detector→pose pipelines (e.g., YOLO-based [4]) communicate directly with multiple people and are generally trained using COCO keypoints [5]. Cross-contrasting of these paradigms with a multi-person image level imposes bias, by means that single-person systems are not trained to match targets with the distractors.

Why this matters. A critical pitfall is the *mismatch between training and evaluation conditions*: single-person models are optimized on cropped person instances, whereas detector-based systems process full images that often contain multiple people. This discrepancy can inject unfair bias and produce misleading conclusions; for example, detector-based pipelines may appear stronger by leveraging multi-person context, while single-person models are penalized for not being trained in such settings.

Goal. To enable a fair comparison, we align test-time inputs with the single person training regime. We put forth a lightweight and reproducible *detector-based fairness protocol* that aligns detected boxes to ground truth via IoU so that all methods are evaluated under the same single-human input condition, eliminating mismatched evaluation effects before pose prediction.

2. Related Work

Single-person HPE. At a multi-scale scale, Hourglass retains context by iterating from bottom to top or from top to bottom [2]. SimpleBaseline provides a residual stem combined with a deconv head that can be very competitive [3].
Multi-person HPE. COCO keypoints established detector-first pipelines as the norm [5]. In zero-shot we utilize YOLOv8-pose and use the Ultralytics interface [6, 7].

3. Method

3.1. Dataset and Metric

MPII. We work with the MPII Human Pose dataset [1] (16 joints).
PCKh. We can follow [1] and say that a joint j is correct if

$$\|\hat{\mathbf{p}}_j - \mathbf{p}_j\|_2 \leq \alpha \cdot \ell_{\text{head}},$$

with $\alpha = 0.5$; we report PCKh@0.5 unless otherwise noted. In the absence of GT head box we have $\ell_{\text{head}} = 0.5 \min(W, H)$.

3.2. Models and Training

Hourglass. Stacked Hourglass with intermediate supervision [2].
SimpleBaseline. A lightweight ResNet-style stem (to $1/32$ size) with a 3-stage deconv head which returns K heatmaps at $1/4$ resolution [3]. Heatmaps are Gaussians ($\sigma = 2$); loss is an Hourglass-style foreground-weighted mean squared error (MSE) (boost 82) for numeric compatibility.

Table 1. Experiment configurations used across all models.

Setting	Value
Input size	256×256
Heatmap size	64×64
Optimizer	AdamW
Weight decay	1×10^{-4}
Gaussian σ	2
Foreground boost	82
IoU threshold	0.2
Visibility usage	Generate targets (no per-joint weights)

YOLOv8-pose. COCO-pretrained `yolov8n-pose` (zero-shot on MPII); we map COCO-17 keypoints to MPII-16 using pelvis/thorax midpoints and a nose proxy for upper-neck/head-top (discussed further below).

3.3. Detector-based Fairness Protocol

We construct a GT box b^{gt} from `objpos/scale` or from padded min–max joint box with an image I. We perform a person detector to find $\{b_i\}$ and select

$$b^\star = \arg\max_i \text{IoU}(b_i, b^{\text{gt}}).$$

If $\max \text{IoU} < \tau_{\text{iou}}$ (using 0.2), we go back to b^{gt}. We crop by given single box and (i) apply crop to Hourglass/SimpleBaseline; (ii) evaluate on same selected person instance for YOLO.

3.4. COCO-17 $\rightarrow$ MPII-16 Mapping

It follows from this point to {`nose, leye, reye, lear, rear, ls, rs, le, re, lw, rw, lh, rh, lk, rk, la, ra`}. We set

$$\text{Pelvis} = (\text{lh} + \text{rh})/2, \quad \text{Thorax} = (\text{ls} + \text{rs})/2,$$

and use `nose` as a surrogate for UpperNeck/HeadTop; limb joints of the body follow index-consistent correlations (e.g., R_Ank $\leftarrow$ ra, R_Knee $\leftarrow$ rk). Coordinates are normalized to the image size.

4. Experiments

4.1. Implementation Details

The input is resized to 256×256; heatmaps are 64×64. Optimizer: AdamW (weight decay 10^{-4}). Visibility is used to produce targets but they are not per-joint loss weights. All experiments share the same configuration, summarized in Tab. 1.

4.2. Evaluation Regimes

- **Regime A (no detector).** Single-person models run on the full image; YOLOv8-pose runs as-is (unfair to single-person models).

Table 2. Mean PCKh@0.5 on MPII (our runs). No detector is used in left column (subset sanity check); right column uses detector-based fairness protocol on the full validation set. A single column width gives the table half-page width.

Model	No-Detector	With-Detector
Hourglass	0.2313	0.6169
SimpleBaseline	0.2272	0.1872
YOLOv8n-pose	0.5600	0.8862

Table 3. Per-joint PCKh@0.5 with detector-based fairness on the full validation set (our runs). Each cell shows PCKh (n). Kept to a single column to fit within half-page width.

Joint	Hourglass	SimpleBase.	YOLOv8n
R_Ank	0.4113 (2115)	0.0028 (2115)	0.8773 (2102)
R_Knee	0.4704 (2485)	0.0044 (2485)	0.8830 (2471)
R_Hip	0.5978 (2889)	0.1063 (2889)	0.9123 (2875)
L_Hip	0.6298 (2888)	0.1236 (2888)	0.9099 (2874)
L_Knee	0.4762 (2478)	0.0077 (2478)	0.8892 (2464)
L_Ank	0.4422 (2119)	0.0028 (2119)	0.8799 (2107)
Pelvis	0.5705 (2878)	0.1751 (2878)	0.9207 (2864)
Thorax	0.7394 (2932)	0.2616 (2932)	0.9311 (2918)
UpperNeck	0.8400 (2932)	0.3233 (2932)	0.8242 (2918)
HeadTop	0.7804 (2932)	0.3987 (2932)	0.7440 (2918)
R_Wrist	0.5400 (2928)	0.2114 (2928)	0.8833 (2914)
R_Elbow	0.5305 (2935)	0.2807 (2935)	0.9011 (2921)
R_Shoulder	0.7772 (2944)	0.3645 (2944)	0.9253 (2930)
L_Shoulder	0.8376 (2944)	0.3040 (2944)	0.9218 (2930)
L_Elbow	0.6412 (2932)	0.2220 (2932)	0.9027 (2918)
L_Wrist	0.5851 (2909)	0.2056 (2909)	0.8736 (2895)

- **Regime B (with detector fairness).** We apply Sec. 3.3 to the full validation set.

4.3. Results (from our runs)

These tables 2 and 3 summarize all results strictly from our experiments.

4.4. Detector Failure: Wrong-Person Assignment

Worst-5 visualizations (GT skeleton: green; GT box: cyan; detector box: orange; selected box: magenta) reveal two common failure patterns:

(1) Extra-person inclusion. The detector sometimes produces boxes that include not only the GT person but also surrounding people. In such cases, the selected box may correspond to another person, so the predicted pose is valid for that person but inconsistent with the GT labels, leading to an apparent failure.

(2) Missed GT detection. Due to detector limitations, the GT person may not be detected at all, resulting in all candidate boxes having maximum IoU < 0.2. In this case, our fallback rule uses the GT box, but YOLO predictions (based only on detected boxes) cannot cover the GT, and per-image accuracy drops to 0. These cases highlight the sensitivity of cross-paradigm evaluation to detector recall. We recommend reporting detector recall at the chosen IoU threshold, and additionally computing PCKh conditioned on successful matches, for clearer interpretation. See Fig. 1.

4.5. Why SimpleBaseline Did Not Improve

Observed drivers: (i) weaker high-resolution pathway than Hourglass; (ii) foreground-weighted MSE without per-joint visibility weights; (iii) tighter detector crops than training distribution; (iv) coarse 64×64 heatmaps without quarter-offset/DARK; (v) crop normalization mismatch. Remedies: longer training with flip/scale/rotation, visibility-weighted losses, crop-aware training, higher heatmap resolution, and DARK decoding.

5. Ablations and Reproducibility

Ablations to try. Sweep IoU threshold $\{0.0, 0.2, 0.5\}$; crop margin 1.0–1.5; heatmap size $64 \rightarrow 96$; add per-joint weights and DARK.

Reproducibility. Resize 256×256, heatmaps 64×64, Gaussian σ=2, foreground boost 82, τ_{iou}=0.2, PCKh@0.5, COCO→MPII mapping in Sec. 3.4.

6. Limitations and Future Work

YOLO is zero-shot on MPII; the nose proxy for upper-neck/head-top is imperfect. We omit multi-scale/flip testing and advanced decoders. Training single-person models on detector-style crops should further clarify gains under Sec. 3.3.

7. Conclusion

In this work, we introduced a detector-based fairness protocol for cross-paradigm human pose estimation comparison on MPII. Our contributions are threefold: (i) we identified and formalized the evaluation mismatch between single-person and detector-based pipelines, (ii) we proposed an IoU-based protocol to ensure fair input alignment, and (iii) we quantified the effect of this protocol through PCKh evaluation and failure analysis.

While the protocol improves fairness, limitations remain: YOLOv8-pose is applied in a zero-shot manner on MPII, and the COCO→MPII keypoint mapping (e.g., using the nose as a proxy for upper-neck/head-top) is imperfect. Future work should retrain single-person models with detector-style crops and extend the protocol to other datasets and metrics.

By reporting both PCKh and detector recall, researchers can avoid confounding effects and achieve more reliable and transparent evaluations in human pose estimation.

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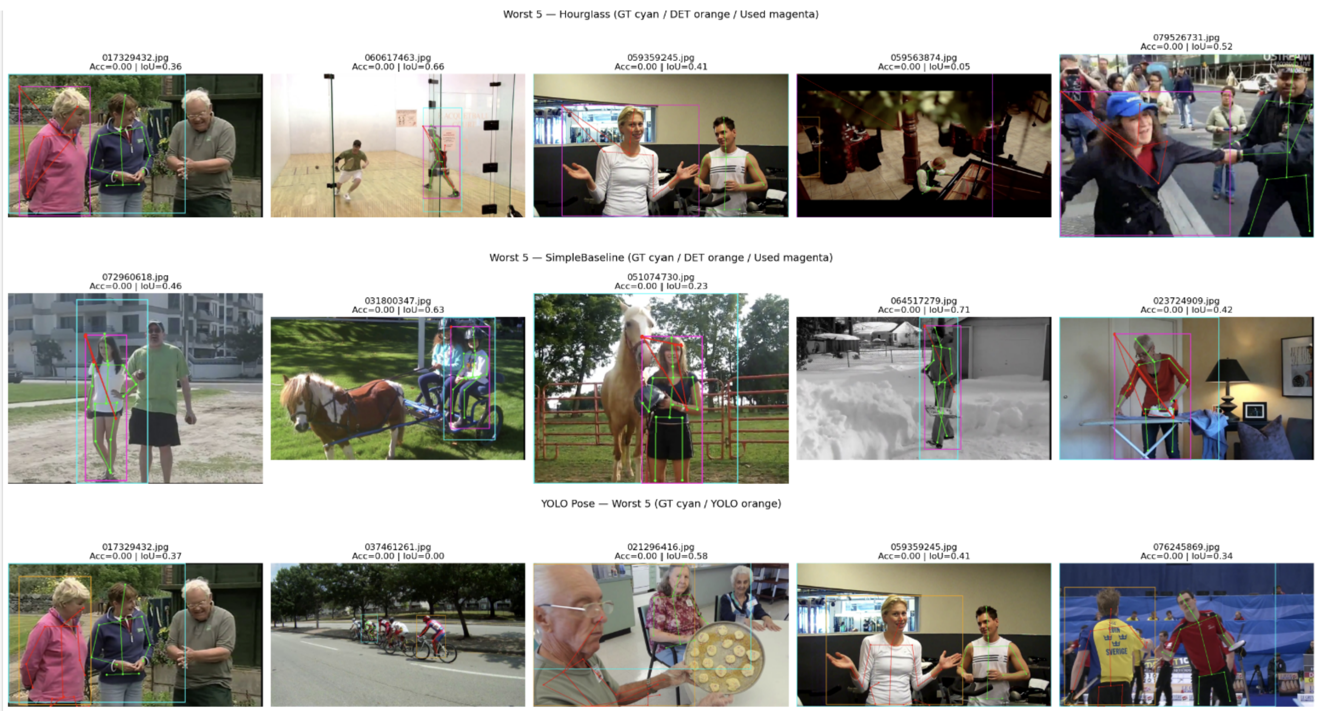

Figure 1. **Worst-5 samples per model.** Rows: Hourglass (top), SimpleBaseline (middle), YOLOv8-pose (bottom). Colors: GT skeleton (green), GT bbox (cyan), detector bbox (orange), selected/used bbox (magenta). When the detector fails to produce a box for the GT person, the pipeline selects a different person, leading to zero PCKh despite a valid pose on another instance.